AI WORKFLOW ORCHESTRATION • CAPSTONE PROJECT

Relay

Design, execute, and recover AI-powered workflows—with control, visibility, and reliability built in.

FROM FRAGILE AUTOMATIONS to **durable, observable workflows**	**AI assists.** **Relay stays in control.**

THE PROBLEM

Workflow automation becomes brittle when steps fail, AI returns malformed or unsafe output, or an external action succeeds just before a process crashes. Without durable state and clear run history, teams struggle to recover safely and trust the result.

THE SOLUTION

Relay is an AI workflow orchestrator that lets teams connect typed steps into workflows, publish controlled versions, trigger runs through an API or webhook, and execute them in the background. It persists progress, validates AI outputs, pauses for human approval when needed, and resumes safely after failures.

BRIEF OVERVIEW

Create workflows as drafts, publish immutable versions, and inspect every run from trigger to outcome. Relay combines deterministic workflow logic with AI-powered steps while keeping permissions, state transitions, and side effects under explicit control.

KEY BENEFITS

- **Resilient execution:** durable jobs, retries, and crash recovery.
- **Safer AI:** schema validation and a bounded repair attempt.
- **Human oversight:** approval gates before sensitive actions.
- **Fewer duplicates:** idempotency keys for external side effects.
- **Clear visibility:** run traces, step attempts, and failure details.
- **Guardrails:** step limits and controlled workflow publishing.

CORE FEATURES

01 · Workflow Builder	Typed nodes and explicit connections define a workflow's execution path.	**02 · Publish & Trigger**	Immutable published versions; manual and webhook triggers.
03 · Durable Worker	Background execution with persisted state, retries, and restart recovery.	**04 · AI + Validation**	Structured AI results checked against schemas before downstream use.
05 · Approval Gates	Pause execution for a person to approve or reject sensitive steps.	**06 · Run Timeline**	Inspect step-by-step inputs, outputs, attempts, and final status.

BUILD WORKFLOWS THAT KEEP MOVING — EVEN WHEN A STEP FAILS.

Relay • AI Workflow Orchestrator | Java + Spring Boot | PostgreSQL | AWS demo deployment